A Unified Multiview Spectral Feature Learning Framework for Hyperspectral Image Classification

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***Abstract*— Recent progress in spectral classification is dominated by the use of deep learning models. While various learning architectures have been developed, they all extract spectral features from a single-view input. In this article, we investigate a different perspective and develop a unified multiview spectral feature learning framework, which extracts discriminative spectral features from multiple views of inputs. To our knowledge, this is the first reported multiview spectral feature learning method based on deep learning. In this framework, we introduce a multiview spectrum construction method by transforming the input spectral vector into multiple 3-D image patches with different sizes, termed multiview spectrum. This multiview spectrum is fed to a well-designed triple-stream architecture, where global and two local spectral feature learning networks operate in parallel, thus capturing both global and local spectral contextual features simultaneously. Another important contribution of this work is a novel interactive attention mechanism to identify the most informative spectral contextual features. The model is trained in an end-to-end fashion from scratch with a joint loss. Experimental results on four datasets demonstrate excellent performance compared to the current state of the art.**

***Index Terms*— Attention mechanism, deep learning, hyperspectral image (HSI) classification, multiview spectrum.**

I. Introduction

HYPERSPECTRAL image (HSI) is composed of both spatial information and spectral information, and each pixel vector encompasses several dozens to hundreds of continuous wavelength channels, which holds tremendous potential for accurate land-cover identification [1]. Hence, various methods for HSI processing have been reported recently [2], [3], [4], [5], [6]. Scene classification remains to be one of the most fundamental and demanding tasks since it is an essential step in a range of applications, e.g., urban mapping [7], precise agriculture [8], military surveillance [9], and ecosystem service [10].

Recent studies demonstrate that spatial information plays a vital role in improving HSI classification [11], [12], [13], [14], [15], [16]. Reasoning that making use of spatial information greatly mitigates the spectral variability, mixed pixels, and other adverse effects in the data, spectral–spatial classification typically outperforms spectral classification alone [17], [18], [19], [20]. In HSI processing, the training and testing sets are taken from the same image, and spectral–spatial classification that operates on 3-D image patches inevitably suffers from overlapping problems between neighboring training and testing samples to a certain extent. Especially when the window size of the input and the amount of training data increase or the spatial scale of HSI is relatively small, this probably leads to exaggerated classification accuracy in the tests and thus may affect a fair evaluation of the methods, as indicated in [21]. This article aims to avoid overlapping while yielding the state-of-the-art classification accuracy.

Spectral classification alone is free from the abovementioned overlapping problem since it takes as input a spectral vector corresponding to a single pixel across all the spectral bands. Consequently, the training and testing sets are mutually exclusive. Spectral classification remains to pose huge challenges due to the very high dimensionality together with the shortage of labeled data to match it [22], [23]. The related phenomena are referred to as the curse of dimensionality [24]. Researchers have proposed a variety of traditional spectral classifiers [25], including support vector machines [26], manifold learning [27], random forests [28], and active learning [29], to deal with various classification problems. Recently, diverse deep learning models, including stacked autoencoder (SAE) [30], deep belief network (DBN) [31], convolutional neural network (CNN) [32], [33], [34], recurrent neural network (RNN) [35], [36], and graph convolutional network (GCN) [37], have been applied to spectral feature extraction. Compared with traditional classifiers, deep learning models enable high-level spectral feature extraction, thereby improving the classification accuracy [38]. However, the classification maps for these spectral classifiers still tend to be noisy due to spectral variability, and the use of spatial information in the postprocessing stage is an effective way [34]. The main idea is to train the model only with spectral information and then to incorporate the spatial information (e.g., voting strategies [39], [40]) without involving the training set at the testing stage as refinement.

Although the above-described deep learning models [30], [31], [32], [33], [34], [35], [36], [37], [41] have demonstrated the success in HSI spectral classification, how to exploit the spectral signatures more effectively remains an

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important challenge. Although deep neural networks are in theory universal approximators (i.e., can learn to approximate practically any function and extract arbitrary features), in practice, we meet performance bottlenecks when running deep neural networks on 1-D spectral vectors. The available training data may not be sufficient to learn extracting most distinctive features in complex scenes. Furthermore, the existing approaches are not capable of capturing both local and global spectral features [37] (either local spectral features learned by CNNs or global spectral features extracted by the other deep learning models), leading to a weaker spectral feature representation. When employing CNN models (e.g., in [32], [33], and [34]), those used 1-D CNN to extract local spectral features from pixel vectors and ignored global spectral features due to its local connection mechanism.

In this article, we propose a novel multiview spectral feature learning framework based on 2-D CNN for HSI classification, which captures simultaneously global and local spectral contextual features. In this framework, inspired by the success of multiview learning [42], [43] and diverse region construction [44], we devise a multiview spectrum construction method that transforms the input spectral vector into multiple 3-D image patches with different sizes. The differently composed 3-D patches serve as multiple views of the spectral input and this way reshaped inputs can be fed directly into powerful 2-D CNN or 3-D CNN models. The benefit from this approach is twofold: 1) we can directly use 2-D (as well as 3-D) CNN models for spectral-alone classification, benefiting from their powerful feature extraction capabilities, and 2) multiple views of the same spectral input facilitate the extraction of the most discriminant features. We then develop a triple-stream spectral feature learning network based on 2-D CNN to make full use of the constructed multiview spectrum data. The first stream aims to extract global spectral correlation features from a view of relatively small image patches. The other two streams perform a multikernel depthwise convolution to extract local spectral contextual features from two views of relatively large image patches. Moreover, we develop a novel interactive attention embedding into the two local streams to further boost the spectral contextual feature learning capability. Finally, all the triple parallel streams are integrated adaptively via a decision fusion. At the testing stage, we make the final prediction via a voting method, which employs spatial information and avoids the overlapping problem.

The main contributions of this article are given as follows.

1) We propose a unified multiview CNN-based spectral feature learning framework for HSI classification. We are not aware of any reported works on multiview spectral feature learning based on deep learning. The main advantage of the proposed framework is its discriminative spectral feature representation power, exhibited as a remarkably improved performance when applied to challenging datasets.
2) We introduce a simple but effective multiview spectrum construction method. This multiview construction method can be used in combination with arbitrary 2-D CNN or 3-D CNN models. Moreover, we develop a triple-stream spectral feature learning network to extract both global and local spectral contextual features from the multiple views of spectrum.
3) We develop a novel interactive attention method for improving the spectral contextual feature learning capability based on global correlations. This attention method can be also applied to other multistream networks.

The rest of this article is organized as follows. Section II reviews related work on spectral classification and attention mechanism. Section III introduces the details of the proposed method. A thorough experimental evaluation and a discussion of the results are given in Section IV, and Section V draws the conclusion of this article.

II. Related Work

A. Spectral Classification

Spectral classification aims to assign a unique label to each pixel vector in an HSI. A range of traditional classification methods have been applied to address various classification problems [25], [45]. The representative models include support vector machines [26], manifold learning [27], random forests [28], and active learning [29]. Recently, diverse deep learning models have been developed, exhibiting great potential in spectral feature extraction and classification.

To date, six types of deep architectures for spectral classification have been proposed, building on SAEs [30], DBNs [31], CNNs [32], [33], [34], RNNs [35], [36], generative adversarial networks (GANs) [41], and GCNs [37]. Both SAE and DBN are fully connected networks where their performances are severely hampered by the number of training data. The CNN models have gained great attention due to their local connection and shared-weight mechanisms, reducing hugely the number of parameters, while these CNN models extract local spectral features [32], [33], [34]. RNNs process spectral information from a sequential perspective, capturing nonadjacent spectral features [35], [36]. The GAN model mitigates the issue of overfitting by generating pseudo samples and adversarial training [41]. The GCN model is capable of generalizing the operation of convolution from grid data to graph data by using a non-Euclidean structure [37]. Representatives of these CNNs [32], [33], [34], RNNs [35], [36], GAN [41], and GCN [37] are chosen for comparison in Section IV-B.

Although the above-described methods demonstrated huge success, they typically employed a single-view input [32], [33], [34], [35], [36], [37], [41] for spectral classification, and when employing CNN (e.g., in [32], [33], and [34]), those used 1-D convolutions to extract local spectral features and none within the attention mechanism. We instead propose a multiview spectral feature learning framework based on 2-D CNN, which captures both global and local spectral contextual features from multiple-view inputs. Moreover, we embed an interactive attention module into the framework to further enhance its representation power.

B. Attention Mechanism

The attention mechanism has been widely used in diverse deep learning models to boost the feature representations [46].

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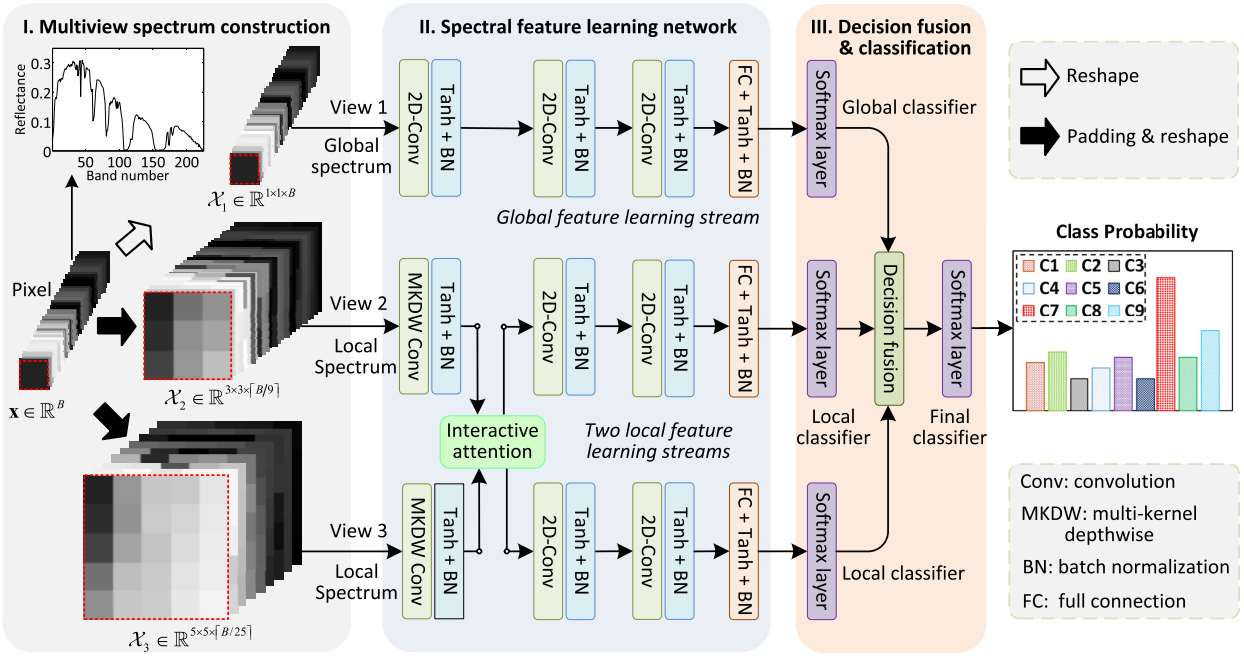


Fig. 1. Overall framework of the proposed method. It comprises a multiview spectral construction, a spectral feature learning network, and a decision fusion that are combined into a unified framework and trained in an end-to-end fashion. The spectral feature learning network is a triple-stream architecture, where a global stream and two local streams operate in parallel, capturing both global and local spectral contextual features from the constructed multiview spectrum data. The interactive attention is embedded into the two local streams to further enhance their feature learning capability.

Hu et al. [47] proposed channel attention to learn the channel-wise correlations from global averaging feature maps. Lately, Wang et al. [48] designed efficient channel attention based on 1-D convolution operation to capture local correlations. Subsequently, Hou et al. [49] introduced a cross-attention module, which models adaptively the semantic correlation to generate more discriminative features for few-shot classification. Very recently, Chen et al. [50] developed an across attention that exchanges information between two branches efficiently to enhance the representation power.

In the HSI classification, the attention mechanism was often utilized to boost the spectral–spatial feature extraction capability. For instance, Mou and Zhu [51] designed a spectral attention method to learn important spectral bands from global spatial information in spectral–spatial classification. Yang et al. [52] proposed a cross-attention spectral–spatial attention network, which consists of cross-spectral attention and cross-spatial attention in series, in order to suppress redundant bands and to fuse spectral–spatial features, respectively. Very recently, Yang et al. [53] developed an adaptive graph cross-attention fusion module to suppress noise interference and to fuse the spectral and spatial features extracted from a dual-branch network.

The attention approaches mentioned above mainly aimed to improve spectral–spatial [51], [52], [53] representation power and often employed the pooling operation [47], [54] or the convolution operation [48], [51] to reduce the model complexity, leading to the information loss. In contrast, we here develop interactive attention to improve spectral feature learning capability with the complete input features, embedding in the two local feature learning networks.

III. Methodology

A. Overall Architecture

We propose a unified multiview spectral feature learning architecture based on 2-D CNN for HSI classification, which captures simultaneously global and local spectral contextual features from multiple views of spectral inputs, with the proposed multiview spectrum construction. Our model is a triple-stream architecture, where a global feature learning stream and two local feature learning streams operate in parallel, as shown in Fig. 1. While the global stream extracts global spectral features, the two local streams learn the local spectral contextual features. The proposed interactive attention method is embedded into the two local streams to improve the local spectral context feature learning capability. The outputs of the triple streams are integrated with a decision fusion method to predict the outputs of labels.

B. Multiview Spectrum Construction

Current deep spectral classifiers [32], [33], [34], [35], [36] typically adopt a single-view input for spectral feature extraction. This leads to a weaker spectral representation power due to the inherent limitations of models and the complex spectral signatures caused by different spatial resolutions together with various land-cover types. Inspired by the success of multiview learning [42], [43] and diverse region construction [44], we here devise a multiview spectrum construction method to provide multiple views of spectral inputs, facilitating this way the extraction of relevant features by a deep learning model. The main idea is to transform the input spectral vector into multiple 3-D image patches with different sizes and then to feed them to a well-designed multistream network.

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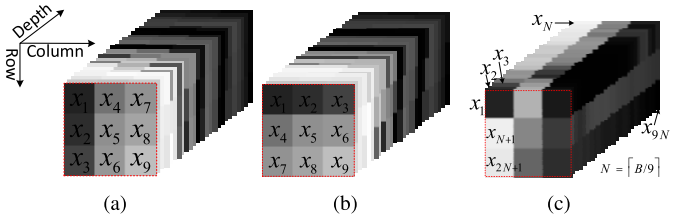


Fig. 2. Illustration of 3-D cube construction from a spectral vector of one pixel $\mathbf{x} \in \mathbb{R}^B$ in three orderings. (a) RCD ordering that the row axis first and then the column axis to the depth axis. (b) CRD ordering that the column axis first and then the row axis to the depth axis. (c) DRC ordering that the depth axis first and then the row axis to the column axis.

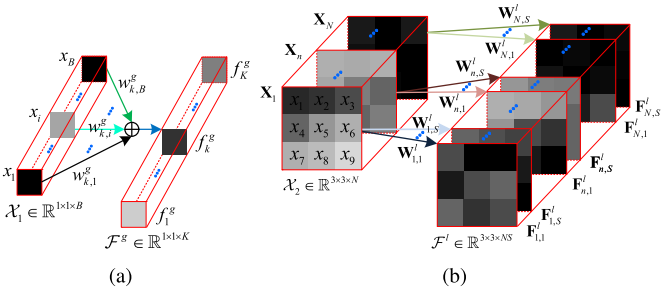


Fig. 3. Illustrations of the proposed (a) global spectral feature extractor and (b) local spectral contextual feature extractor. Different colors of the arrows represent here different weights. $\oplus$ denotes the addition operation.

The initial input is a spectral vector composed of pixel values in all spectral bands at a particular spatial location. We refer to it as one pixel $\mathbf{x} \in \mathbb{R}^B$, where B is the number of bands. We transform $\mathbf{x}$ into an image patch $\mathcal{X} \in \mathbb{R}^{P \times P \times N}$. For a desired P, we set $N = \lceil B/P^2 \rceil$, where $\lceil \cdot \rceil$ denotes the ceil function, and the input vector is brought to the size P^2N by zero padding. We do so to construct multiview spectrum data by setting different P values. In our concrete implementation, we set P to 1, 3, and 5. In this way, one view is the original 1-D spectral vector as in traditional spectral classification methods $\mathcal{X}_1 \in \mathbb{R}^{1\times1\times B}$, and two other views are $\mathcal{X}_2 \in \mathbb{R}^{3\times3\times\lceil B/9\rceil}$ and $\mathcal{X}_3 \in \mathbb{R}^{5\times5\times\lceil B/25\rceil}$, as shown in the left of Fig. 1. We refer to this approach as multiview spectrum construction because the resulting 3-D patches can be seen as multiple views of the same spectral input. This method offers a different perspective that processes spectral information from 3-D image patches instead of from traditional 1-D vectors. A key advantage is to build different views of 3-D spectral inputs, which allows powerful 2-D CNNs or 3-D CNNs to learn discriminative (e.g., global and local contextual) spectral features from them.

Multiview spectrum construction naturally raises the question: how to choose the index ordering of each spectral vector for multiview spectrum construction appropriately? Would it affect the final classification performance of the proposed framework? Fig. 2 shows three possible orderings of 3-D cube construction from a spectral vector, i.e., RCD ([row][column][depth]), CRD ([column][row][depth]), and DRC ([depth][row][column]). Since the proposed spectral feature learning network is based on 2-D CNN, we select RCD or CRD ordering to generate the spectral context in the first two dimensions. More evidence can be found in Section IV-C. Note that the proposed multiview spectrum construction approach is simple but effective, and it can be applied to other 2-D CNN or 3-D CNN models.

C. Spectral Feature Learning Network

Taking full advantage of the constructed multiview spectrum data, we develop a triple-stream spectral feature learning network based on 2-D CNN to capture discriminative spectral features. The main idea is to encode the multiview spectrum inputs into their global and local contextual representations, with a well-designed triple-stream 2-D CNN model in order to process spectral features at different views in parallel. In this way, we aim to extract both global and local spectral contextual features simultaneously. The middle of Fig. 1 shows the proposed spectral feature learning network, which consists of a global feature learning stream and two local ones.

1) Global Feature Learning Stream: The global feature learning stream in our architecture employs a 2-D CNN to capture global spectral correlation features. The input is a view of 3-D image patch extending over all the spectral bands at the third depth dimension and containing thus global spectral information. While earlier reported spectral feature learning methods were always based on 1-D vectors and extracted local spectral features [32], [33], [34] or nonadjacent spectral features [35], [36], our method captures global spectral correlation features via a regular 2-D CNN. Fig. 3(a) shows this architecture.

Let $\mathbf{x} \in \mathbb{R}^B = [x_1, x_2, \ldots, x_B]^T$ denote a spectral vector of one pixel with B bands and transform it into a 3-D image patch $\mathcal{X}_1 \in \mathbb{R}^{1\times1\times B}$. We define the global spectral feature value f_k^g in the kth feature map of the first layer as follows:

$$f_k^g = \delta\left(\mathcal{W}_k^g * \mathcal{X}_1\right) = \delta\left(\sum_{i=1}^{B} w_{k,i}^g \cdot x_i\right) \tag{1}$$

where $\mathcal{W}_k^g \in \mathbb{R}^{1\times1\times B} = [w_{k,1}^g, w_{k,2}^g, \ldots, w_{k,B}^g]$ is the 2-D kernel connected to the kth feature map, $*$ is the convolution operation, and δ is the tanh activation function. The bias terms are omitted in this article. Equation (1) shows that each global spectral feature f_k^g is extracted from whole spectral information $\mathcal{X}_1$ with $\mathcal{W}_k^g$. We employ multiple kernels to extract different global spectral features. The output of the first layer for the global stream is $\mathcal{F}^g \in \mathbb{R}^{1\times1\times K} = [f_1^g, f_2^g, \ldots, f_K^g]$, where K is the number of kernels. We cascade two convolutional layers and one fully connected layer to learn deep global spectral features. For simplicity, let the number of kernels be the same for all the convolutional layers (i.e., K).

2) Local Feature Learning Streams: The two local feature learning streams in our model aim to effectively extract discriminative spectral contextual features. The inputs are two views of 3-D image patches (i.e., $\mathcal{X}_2$ and $\mathcal{X}_3$), which extend over adjacent bands at each channel and contain thus local spectral contextual information, as shown in Fig. 2(b). While current spectral classifiers [32], [33], [34] typically take as input 1-D spectral vectors, and when employing CNN, those exploited 1-D convolution with shared weight to extract local

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spectral features from adjacent bands. However, those methods omit the differences among different portions of spectral bands.

We develop a contextual feature learning method to capture discriminative local spectral contextual features. The main idea is to exploit 2-D multikernel depthwise convolution [55] to expand each input channel (spectral contextual information) into its multiscale representations and then to use two regular convolutional layers to integrate them subsequently. In this way, we aim to formulate spectral contextual features at multiple scales simultaneously, thereby boosting the representation power by considering the differences among different spectral bands. We then extend this idea into multiple views of spectral inputs to extract more discriminative spectral features.

For each stream, let $\mathcal{X} \in \mathbb{R}^{P\times P\times N} = [\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_N]$ denote the input with window size of $P \times P$ and with N channels, where $\mathbf{X}_n \in \mathbb{R}^{P\times P}$ is the nth channel of $\mathcal{X}$. We define the local spectral contextual feature map $\mathbf{F}^l_{n,s} \in \mathbb{R}^{P\times P}$ of the sth scale on $\mathbf{X}_n$ in the first layer as follows:

$$\mathbf{F}^l_{n,s} = \delta\left(\mathbf{W}^l_{n,s} * \mathbf{X}_n\right) \tag{2}$$

where $\mathbf{W}^l_{n,s} \in \mathbb{R}^{3\times 3}$ is the kernel matrix connected to $\mathbf{F}^l_{n,s}$. The corresponding multiscale feature maps $\mathcal{F}^l_n \in \mathbb{R}^{P\times P\times S} = [\mathbf{F}^l_{n,1}, \mathbf{F}^l_{n,2}, \ldots, \mathbf{F}^l_{n,S}]$ are extracted from each local spectral contextual information $\mathbf{X}_n$ with a multikernel depthwise convolution $\mathcal{W}^l_n \in \mathbb{R}^{3\times 3\times S} = [\mathbf{W}^l_{n,1}, \mathbf{W}^l_{n,2}, \ldots, \mathbf{W}^l_{n,S}]$. The output of the first layer for each local stream is $\mathcal{F}^l \in \mathbb{R}^{P\times P\times \mathrm{NS}} = [\mathcal{F}^l_1, \mathcal{F}^l_2, \ldots, \mathcal{F}^l_N]$. Similar to the global stream, we employ two convolutional layers following one fully connected layer to fuse the extracted multiscale features. To reduce the number of hyperparameters, we let the number of kernels and neurons be the same for all the triple streams.

In contrast to [55], which introduces a pointwise multikernel depthwise convolution to weight the spatial information at different scales and aims to extract global spectral features, we here exploit a spatial multikernel depthwise convolution to weight the spectral information at different scales and aims to extract local spectral contextual features.

D. Interactive Attention

We aim to further boost the spectral feature learning capability of the two aforementioned local streams by embedding an interactive mechanism into both of them (see the middle bottom of Fig. 1). We develop a novel interactive attention method based on global correlations. Fig. 4 shows this architecture. Next, we elaborate on its details.

To establish the interactive global correlations, we embed the raw spectral features from the first layers of the two local streams. This is achieved by using flattening and concatenating operations to generate global statistics. Let $\mathcal{F}_1 \in \mathbb{R}^{H_1\times W_1\times C_1}$ and $\mathcal{F}_2 \in \mathbb{R}^{H_2\times W_2\times C_2}$ be the two inputs, and we first vectorize them into corresponding statistics $\mathbf{z}_1 \in \mathbb{R}^{H_1W_1C_1}$ and $\mathbf{z}_2 \in \mathbb{R}^{H_2W_2C_2}$. We then concatenate $\mathbf{z}_1$ and $\mathbf{z}_2$ into the global statistics $\mathbf{z} = [\mathbf{z}_1^T, \mathbf{z}_2^T]^T \in \mathbb{R}^M$ (i.e., $M = \sum_{i=1}^{2} H_iW_iC_i$). In contrast to most of the attention methods, including [47], [48], and [54], which often adopt some form of squeeze operation [such as global average pooling (GAP)] at the input and lead to information loss, our attention method effectively uses the raw spectral features to construct global statistics. This choice preserves more detailed contextual features and facilitates the following feature learning.

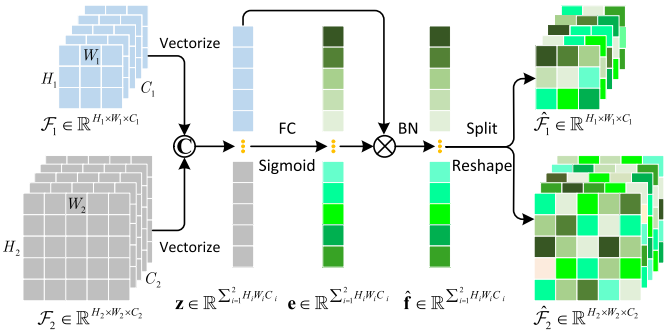


Fig. 4. Diagram of the proposed interactive attention. © illustrates the concatenation. FC represents the fully connected layer. ⊗ denotes the elementwise product. BN is the batch normalization for enhancing gradient stability and facilitating training.

To make use of the constructed global statistics, we opt to introduce a simple gating mechanism that consists of a fully connected layer and the sigmoid activation, which self-learns the interactive contextual correlation vector $\mathbf{e} \in \mathbb{R}^M = [e_1, e_2, \ldots, e_M]^T$. We define the correlation vector as follows:

$$\mathbf{e} = \sigma(\mathbf{W}\cdot\mathbf{z}) \tag{3}$$

where σ refers to the sigmoid function, $\cdot$ denotes the dot product ($\mathbf{x}\cdot\mathbf{y} = \sum_i x_i y_i$), and $\mathbf{W} \in \mathbb{R}^{M\times M} = [\mathbf{w}_1^T, \mathbf{w}_2^T, \ldots, \mathbf{w}_M^T]^T$ is the weight matrix of the fully connected layer. To limit model complexity and aid generalization, we use one fully connected layer with M neurons to automatically learn $\mathbf{e}$ from $\mathbf{z}$. Considering that each element $e_i \in \mathbf{e}$ is extracted from global statistics (i.e., all local contextual features) $\mathbf{z} = [z_1, z_2, \ldots, z_M]^T$ with $\mathbf{w}_i \in \mathbb{R}^M$ that $e_i = \sigma(\mathbf{w}_i^T \cdot \mathbf{z})$, our attention learns thereby global interactive contextual correlations.

To emphasize informative spectral contextual features and suppress less useful ones, we employ the learned correlations to recalibrate the global statistics as follows:

$$\hat{\mathbf{f}} = \mathbf{e}\odot\mathbf{z} \tag{4}$$

where $\odot$ denotes the elementwise product. According to the sizes of $\mathbf{z}_1$ and $\mathbf{z}_2$, we split the recalibrated features $\hat{\mathbf{f}} \in \mathbb{R}^M$ into two feature vectors, $\hat{\mathbf{f}}_1 \in \mathbb{R}^{H_1W_1C_1}$ and $\hat{\mathbf{f}}_2 \in \mathbb{R}^{H_2W_2C_2}$, and then reshape, respectively, them as the final outputs of the interactive attention: $\hat{\mathcal{F}}_1 \in \mathbb{R}^{H_1\times W_1\times C_1}$ and $\hat{\mathcal{F}}_2 \in \mathbb{R}^{H_2\times W_2\times C_2}$. Note that each feature in $\hat{\mathcal{F}}_1$ and $\hat{\mathcal{F}}_2$ is weighted by the corresponding global interactive correlation in $\mathbf{e}$, enhancing the feature learning capacity of the two local streams. The proposed interactive attention can be applied to other multistream networks too.

E. Decision Fusion and Classification

Having extracted global and local spectral contextual features, we introduce a decision fusion method to integrate them respectively (see the right of Fig. 1). We first use the softmax

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layer to predict the probabilities of each class for all the triple streams. Let $\mathbf{p}_i \in \mathbb{R}^C$ be the output probabilities of the ith classifier. We employ an elementwise minimum and an elementwise averaging to fuse them as follows:

$$\mathbf{p}_{\min}[j] = \min_i \mathbf{p}_i[j], \quad \mathbf{p}_{\text{avg}}[j] = \frac{1}{3}\sum_{i=1}^{3} \mathbf{p}_i[j] \tag{5}$$

and then, $\mathbf{p}_{\min} \in \mathbb{R}^C$ and $\mathbf{p}_{\text{avg}} \in \mathbb{R}^C$ are concatenated and weighted into the fused probabilities $\mathbf{p} \in \mathbb{R}^{2C}$ as follows:

$$\mathbf{p} = \mathbf{e}_f \odot \left(\mathbf{p}_{\min} || \mathbf{p}_{\text{avg}}\right) \tag{6}$$

where $||$ denotes the concatenating operation and $\mathbf{e}_f \in \mathbb{R}^{2C}$ is the importance weights that learn automatically from $(\mathbf{p}_{\max}||\mathbf{p}_{\text{avg}})$ with a softmax attention, i.e., a fully connected layer with the softmax function. The fused probabilities are fed into a softmax layer with C nodes (classes) to predict the final classification label. To facilitate the training, we utilize a joint loss with four cross-entropy losses for the final, global, and two local classifiers, to train the whole network in an end-to-end fashion from scratch.

To further improve the spectral classification results, two common spatial postprocessing strategies are utilized: probabilistic graphical models [56], [57] and majority voting [34], [39], [40]. Probabilistic graphical models can exploit spatial contextual information effectively, but they may not be suited to large-scale HSI scenes due to their high memory requirements. Majority voting is based on the fact that adjacent pixels belong to the same class with a high probability. Compared with single-scale voting [34], multiscale voting [39], [40] is more robust due to complex spatial context in hyperspectral data. We adopt a local and global voting [40] at the testing stage since it makes use of both local information and global spatial information to adapt to different data. In addition, it avoids the overlapping problem by excluding all training pixels and their labels before voting. To guarantee the integrity of HSI and to make use of the local spatial information, each excluded training pixel is replaced with the average of its four adjacent unlabelled pixels.

IV. Experimental Results and Analysis

We perform experiments on four hyperspectral datasets: the University of Pavia (denoted as PaviaU), Salinas, Houston2013,[1] and Houston2018.[2] Three objective metrics, overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ), are used for evaluation. For each experiment, we report the mean and standard deviation of the classification results over ten runs with randomly selected training samples.

A. Data Description and Hyperparameter Setting

The PaviaU image, acquired by the ROSIS-03 sensor over an urban area surrounding the University of Pavia, Italy, consists of 610×340 pixels with $C = 9$ classes and $B = 103$ spectral bands covering the spectral range from 0.43 to 0.86 μm with a spatial resolution of 1.3 m.

[1]Available online: https://hyperspectral.ee.uh.edu/?page_id = 459

[2]Available online: https://hyperspectral.ee.uh.edu/?page_id = 1075

TABLE I

Hyperparameters of the Proposed Method for All the Test Images. P Is the Size of the First Two Axes of the Network Inputs. K and L Refer to the Number of Convolutional Kernels and Layers for All the Triple Streams, Respectively. S Denotes the Number of Scales for the Two Local Streams. J Is the Number of Fully Connected Neurons for All the Triple Streams

Hyperparameters	P	K	S	L	J
Value	1/3/5	384	7	3	128

The Salinas image, collected by the AVIRIS sensor over the area of Salinas Valley, CA, USA, has 512×217 pixels with $C = 16$ classes and with $B = 224$ spectral bands covering the spectral range from 0.4 to 2.5 μm with a spatial resolution of 3.7 m. The Houston2013 image, gathered by the Compact Airborne Spectrographic Imager (CASI) over the campus of the University of Houston and the neighboring urban area, contains 349×1905 pixels with $C = 15$ classes and $B = 144$ spectral bands covering the spectral range from 0.38 to 1.05 μm with a spatial resolution of 2.5 m. The Houston2018 image, captured by the CASI 1500 over the area of the University of Houston, is composed of 601×2384 pixels with $C = 20$ classes and $B = 48$ spectral bands covering the spectral range from 0.38 to 1.05 μm with a spatial resolution of 1 m.

In PaviaU and Salinas, we randomly select 50 labeled pixels per class for training, and the remaining labeled pixels are used as the test set to evaluate the classification performance. In Houston2013 and Houston2018, we follow the sample partitioning method as in [58] since it effectively isolates from the training and testing sets with region shape. Note that the training set is totally excluded at the testing stage following [40] for all the test images, avoiding the overlapping problem; 10% randomly selected training samples are used as the validation set to determine the hyperparameters. The network hyperparameters of the proposed method are the same for all the test images and are given in Table I. The number of training epochs is empirically set to 80. The initial learning rate is set to 0.6 for the PaviaU and Houston2018 images and 0.1 for the other two images, and it reduces with a clever strategy.[3] Due to limited training data, we use a data augmentation [33] to increase them by a factor of 4.

B. Comparisons With the State of the Art

We compare the classification performance of the proposed method (dubbed by MCNNS) with the following state-of-the-art deep learning-based methods. The reference methods are divided into five groups.

1) *CNN-Based Spectral Classifiers:* 1-D CNN with a large reception field (LCNN) [32] and 1-D CNN with a deep network (DCNN) [33].
2) *RNN-Based Spectral Classifiers:* RNN with a band grouping strategy (GRNN) [35] and cascaded RNN (CRNN) [36];

[3]https://keras.io/api/callbacks/reduce_lr_on_plateau/

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TABLE II

COMPARISON OF THE CLASSIFICATION ACCURACIES BETWEEN THE PROPOSED METHOD AND THE BASELINES USING THE PAVIAU IMAGE

Classes	Train/Test	LCNN	DCNN	DGAN	CRNN	**MCNN**	PPCNN	NLGCN	ANNA	**MCNNS**
Asphalt	50/6581	78.32±4.18	74.73±1.79	66.72±4.69	73.39±2.79	86.33±2.33	**97.41±1.58**	93.51±1.28	92.36±3.81	96.63±1.93
Meadows	50/18599	76.71±4.78	81.08±2.44	72.71±4.47	64.28±10.54	91.02±3.74	97.00±1.30	92.59±1.71	95.38±3.52	**97.96±3.13**
Gravel	50/2049	81.62±4.20	76.03±4.33	47.50±12.38	69.62±15.33	84.36±2.87	83.53±4.22	91.30±2.65	91.70±4.51	**93.15±4.24**
Trees	50/3014	91.55±4.70	91.38±3.96	93.72±2.04	92.48±1.65	92.48±2.44	80.34±7.58	**95.79±0.94**	94.49±0.99	90.99±2.63
Metal sheets	50/1295	99.50±0.24	99.85±0.12	99.89±0.05	99.54±0.19	99.71±0.14	99.76±0.44	99.90±0.15	**100±0**	**100±0**
Bare soil	50/4979	77.50±5.47	86.74±4.88	82.29±6.41	68.82±7.53	88.55±3.08	73.65±4.48	96.46±1.28	**99.97±0.08**	99.71±0.59
Bitumen	50/1280	90.40±6.03	88.73±3.07	87.95±7.20	90.34±2.68	91.95±1.66	86.25±7.61	94.22±3.31	97.50±2.12	**99.46±0.53**
Bricks	50/3632	81.68±4.20	74.49±1.88	85.64±3.46	78.06±7.83	78.69±4.54	86.51±6.21	91.55±1.37	89.45±6.44	**94.91±2.87**
Shadows	50/897	**99.88±0.09**	99.87±0.16	99.83±0.17	99.81±0.10	99.69±0.12	96.98±2.93	99.74±0.14	96.63±0.73	98.68±0.43
AA(%)	-	86.35±1.23	85.88±1.00	76.16±2.76	81.82±1.85	90.31±0.36	89.05±1.67	95.00±0.44	95.28±0.71	**96.83±0.80**
OA(%)	-	80.38±2.07	81.89±1.18	81.81±2.55	72.30±4.36	89.20±1.50	90.12±1.56	93.69±0.78	94.93±1.70	**97.09±1.61**
$\kappa \times 100$	-	74.86±2.43	76.69±1.41	69.90±3.31	65.30±4.60	85.83±1.85	87.14±1.99	91.72±1.00	93.33±2.19	**96.15±2.09**

TABLE III

COMPARISON OF THE CLASSIFICATION ACCURACIES BETWEEN THE PROPOSED METHOD AND THE BASELINES USING THE SALINAS IMAGE

Classes	Train/Test	LCNN	DCNN	DGAN	CRNN	**MCNN**	PPCNN	NLGCN	ANNA	**MCNNS**
Brocoli Weeds_1	50/1959	98.36±0.70	99.03±0.48	97.90±2.96	95.59±3.03	99.73±0.12	99.88±0.29	99.35±0.71	**100±0**	**100±0**
Brocoli Weeds_2	50/3676	98.88±0.47	99.02±0.76	99.58±0.32	97.20±4.43	99.86±0.09	99.44±0.31	99.74±0.37	**100±0**	**100±0**
Fallow	50/1926	94.70±6.05	97.04±1.69	85.36±8.29	93.83±6.69	99.40±0.47	95.49±3.42	99.72±0.26	**100±0**	**100±0**
Fallow_plow	50/1344	99.53±0.37	99.46±0.23	99.60±0.12	99.27±0.56	99.58±0.23	96.61±1.27	98.24±0.49	99.82±0.17	**99.93±0.08**
Fallow_smooth	50/2628	94.79±2.83	94.98±1.07	97.29±0.64	97.68±0.49	98.79±0.40	99.03±1.10	99.35±0.06	98.92±0.63	**99.57±0.29**
Stubble	50/3909	98.96±0.86	98.94±0.54	99.63±0.05	99.49±0.22	99.81±0.06	99.86±0.07	99.78±0.19	**100±0**	**100±0**
Celery	50/3529	99.35±0.21	99.47±0.12	99.70±0.05	99.23±0.40	99.59±0.12	99.32±1.41	98.83±0.26	**100±0**	**100±0**
Grapes_untrained	50/11221	67.9±16.59	64.02±8.96	64.23±2.86	59.80±11.91	74.73±5.09	84.82±2.06	83.40±3.44	**87.35±3.76**	87.21±5.90
Soil vinyard	50/6153	96.42±1.63	98.59±0.70	99.70±0.07	97.14±3.85	99.68±0.30	99.15±0.35	97.43±0.30	99.99±0.02	**100±0**
Corn weeds	50/3228	87.96±3.68	90.93±1.54	70.94±4.70	84.96±4.89	94.77±1.18	87.71±3.45	92.63±1.80	97.68±1.31	**97.97±0.86**
Lettuce_4wk	50/1018	95.97±5.38	95.76±1.90	90.28±2.18	95.19±2.30	99.10±1.13	88.55±7.96	97.38±1.97	**100±0**	99.90±0.20
Lettuce_5wk	50/1877	99.37±0.80	99.89±0.12	99.99±0.02	97.82±2.22	99.92±0.09	98.60±1.02	**100±0**	**100±0**	**100±0**
Lettuce_6wk	50/866	97.08±2.23	97.85±0.78	97.41±0.38	96.54±1.39	98.88±0.55	98.36±1.67	**99.65±0.49**	**99.65±0.40**	99.52±0.36
Lettuce_7wk	50/1020	94.99±2.81	94.53±1.50	91.08±0.99	92.68±1.34	97.62±1.26	90.98±7.67	98.43±0.89	**99.95±0.12**	99.55±0.41
Vinyard_untrained	50/7218	63.73±13.50	70.47±5.29	65.86±5.52	67.46±8.59	73.75±3.68	71.99±6.15	81.92±5.11	84.56±7.26	**86.20±5.90**
Vinyard_trellis	50/1757	98.04±1.36	98.35±0.48	95.75±1.36	96.15±1.47	98.96±0.42	98.55±0.95	98.56±0.52	99.74±0.25	**99.98±0.03**
AA(%)	-	92.88±1.04	93.65±0.57	90.89±0.55	91.88±0.98	95.88±0.26	94.27±1.19	96.53±0.31	97.98±0.51	**98.11±0.41**
OA(%)	-	86.17±2.17	86.86±1.63	84.69±0.41	84.61±1.71	90.50±0.74	90.97±1.42	92.96±0.47	95.04±1.20	**95.28±1.18**
$\kappa \times 100$	-	84.63±2.36	85.41±1.78	83.04±0.46	82.93±1.85	89.43±0.81	89.96±1.57	92.16±0.52	94.47±1.34	**94.74±1.31**

TABLE IV

COMPARISON OF THE CLASSIFICATION ACCURACIES BETWEEN THE PROPOSED METHOD AND THE BASELINES USING THE HOUSTON2013 IMAGE

Classes	Train/Test	LCNN	DCNN	GRNN	CRNN	**MCNN**	PPCNN	CAE	**MCNNS**
Healthy grass	198/1053	82.97±0.07	82.91±0.17	**96.95±1.23**	82.53±0.25	82.76±0.09	83.09±0.03	81.58±0.74	82.99±0.03
Stressed grass	190/1064	83.64±0.16	82.61±0.83	**95.53±3.64**	83.03±0.33	83.90±0.13	83.58±0.42	82.85±1.22	83.31±0.36
Synthetic grass	192/505	99.60±0.13	99.98±0.06	71.79±18.16	99.68±0.13	99.98±0.06	**100±0**	66.55±10.39	**100±0**
Trees	188/1056	90.96±0.30	90.01±0.71	93.66±6.66	91.50±0.71	93.41±2.44	92.82±0.15	91.74±0.78	**94.34±2.17**
Soil	186/1056	98.77±0.21	97.89±0.38	96.13±1.49	97.27±0.60	99.19±0.13	99.91±0.07	96.13±1.21	**100±0**
Water	182/143	97.41±1.82	95.73±1.27	45.57±32.46	**98.95±0.72**	95.38±0.84	96.50±2.83	91.12±6.43	85.45±0.28
Residential	196/1072	**87.87±0.92**	83.58±1.44	66.31±14.24	71.53±2.27	86.70±1.37	87.07±3.69	86.40±2.53	86.22±1.20
Commercial	191/1053	41.42±1.06	36.87±1.14	73.80±3.34	36.44±1.07	75.41±2.53	44.00±2.18	62.88±5.01	**78.43±1.55**
Road	193/1059	80.09±1.11	75.05±2.28	81.25±5.11	67.44±1.51	81.62±1.96	81.44±0.97	83.64±1.61	**89.64±1.97**
Highway	191/1036	45.75±2.00	51.92±2.80	**83.34±4.84**	56.87±1.99	75.27±9.05	50.87±2.67	50.74±5.58	80.82±12.97
Railway	181/1054	75.96±1.89	77.41±1.99	88.20±1.98	77.03±1.32	88.54±2.61	82.07±0.63	69.90±2.95	**90.44±3.12**
Parking lot 1	192/1041	68.95±2.26	70.81±3.60	76.71±2.74	64.05±2.12	94.77±0.59	83.02±2.43	92.07±2.55	**98.90±0.53**
Parking lot 2	184/285	71.16±1.99	66.84±2.25	34.61±2.34	67.19±2.06	80.25±1.20	83.33±1.59	**86.70±2.30**	64.91±1.92
Tennis court	181/247	98.95±0.27	99.60±0.26	92.31±1.93	99.43±0.32	98.99±0.27	99.96±0.12	97.57±3.38	**100±0**
Running track	187/473	98.48±0.16	98.16±0.14	99.43±0.62	97.12±1.28	98.22±0.17	99.13±0.45	59.75±14.4	**100±0**
AA(%)	-	81.47±0.39	80.62±0.65	79.71±2.12	79.34±0.35	88.96±0.60	84.45±0.40	79.97±1.24	**89.03±0.87**
OA(%)	-	78.23±0.31	77.47±0.84	78.96±0.64	75.59±0.35	87.44±0.76	81.26±0.45	79.17±0.92	**89.07±1.11**
$\kappa \times 100$	-	76.41±0.34	75.66±0.90	77.27±0.67	73.79±0.38	86.36±0.83	79.80±0.48	77.47±0.99	**88.13±1.20**

3) *GAN-Based Spectral Classifier:* Dual-channel capsule GAN (DGAN) [41].
4) *Deep Classifiers With Spatial Postprocessing:* Pixel-pair CNN model with a majority voting (PPCNN) [34] and artificial neural network with an adaptive majority voting strategy (ANNA) [39].
5) *Spectral–Spatial Classifiers:* A nonlocal GCN (NLGCN) [59] and CAE [58].

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TABLE V

COMPARISON OF THE CLASSIFICATION ACCURACIES BETWEEN THE PROPOSED METHOD AND THE BASELINES USING THE HOUSTON2018 IMAGE

Classes	Train/Test	LCNN	DCNN	GRNN	CRNN	**MCNN**	PPCNN	CAE	**MCNNS**
Healthy grass	1458/8341	68.34±4.59	71.69±7.75	76.98±14.83	60.91±10.80	**83.32±2.43**	77.53±6.70	47.27±6.75	72.84±2.82
Stressed grass	4316/28186	**87.78±0.71**	87.09±0.80	83.92±4.29	76.32±1.53	82.99±1.08	83.52±2.18	83.99±1.59	81.03±1.12
Synthetic grass	331/353	**100±0**	**100±0**	64.73±20.77	99.46±0.46	**100±0**	**100±0**	85.75±8.86	**100±0**
Evergreen trees	2005/11583	94.91±0.42	91.43±0.87	84.13±2.69	87.71±1.51	86.51±0.55	93.64±0.57	**95.89±0.87**	86.79±0.68
Deciduous trees	676/4372	**57.79±2.92**	53.18±2.86	40.38±9.78	38.70±2.75	36.21±1.41	55.29±5.68	51.75±3.79	33.72±1.69
Soil	1757/2759	28.56±0.76	30.30±1.18	**41.77±14.84**	26.35±0.58	28.01±0.25	27.96±3.01	32.25±3.17	30.02±0.42
Water	147/119	22.35±11.36	**41.26±23.51**	11.44±18.03	29.92±0.56	25.13±7.00	29.41±1.68	31.93±4.80	20.50±13.48
Residential	3809/35953	76.89±0.50	75.19±0.61	46.16±3.23	72.59±2.51	76.93±0.38	76.49±1.27	**90.43±1.87**	83.85±0.36
Commercial	2789/220895	41.76±5.17	43.51±2.73	**89.18±1.97**	39.87±5.07	67.52±3.30	52.93±4.92	62.99±3.76	74.37±3.32
Road	3188/42622	45.58±1.15	42.07±0.52	40.68±2.29	38.66±1.74	49.04±0.42	39.91±3.01	**59.27±5.45**	48.66±0.59
Sidewalk	2699/31303	44.77±1.88	49.13±1.44	22.88±2.97	50.41±2.13	66.26±0.48	49.64±4.67	59.00±2.50	**74.46±0.38**
Crosswalk	255/1291	0±0	0.55±0.30	4.27±0.83	**7.94±0.97**	4.35±0.52	2.53±2.99	5.20±1.86	0.0±0.0
Major thorough-fares	5193/41165	51.68±0.59	55.69±1.04	42.02±4.91	47.79±1.53	48.83±0.93	47.80±5.73	53.32±5.10	**56.42±1.62**
Highway	700/9149	51.98±3.03	35.52±1.62	**59.95±3.01**	32.68±4.00	37.62±0.78	43.65±8.51	50.21±10.15	38.84±0.86
Railway	1224/5713	80.58±2.73	80.94±2.65	**86.74±4.81**	54.44±1.77	58.25±1.08	65.13±7.76	61.77±6.12	58.49±0.65
Paved parking lot	1179/10296	75.49±1.66	71.70±2.23	65.68±6.42	71.09±2.50	70.55±1.05	74.72±2.68	**85.38±3.13**	69.19±1.40
Gravel parking lot	127/22	26.36±27.93	**100±0**	11.36±3.86	99.55±1.36	99.55±1.36	**100±0**	98.18±4.17	**100±0**
Cars	848/5730	70.51±3.63	68.46±2.27	19.29±2.83	63.42±3.79	77.10±0.91	77.30±4.19	67.46±6.66	**93.27±0.64**
Trains	493/4872	67.30±3.75	66.06±1.87	24.56±6.42	72.36±2.62	82.97±0.73	61.04±9.31	**94.83±2.33**	89.04±0.79
Seats	1313/5511	**70.29±7.64**	42.51±3.59	35.19±5.33	48.64±2.31	58.77±2.02	25.68±2.67	66.18±3.42	59.66±1.78
AA(%)	-	58.15±1.36	60.31±1.18	47.56±2.36	55.94±0.81	61.99±0.46	59.21±1.27	**64.15±1.49**	63.56±0.72
OA(%)	-	52.75±2.43	52.92±1.19	53.17±2.64	48.71±2.68	65.38±1.56	56.43±2.98	65.65±1.45	**70.25±1.57**
$\kappa \times 100$	-	44.89±2.13	44.76±1.03	44.14±2.38	40.10±2.42	56.64±1.59	47.52±2.86	57.72±1.38	**62.26±1.67**

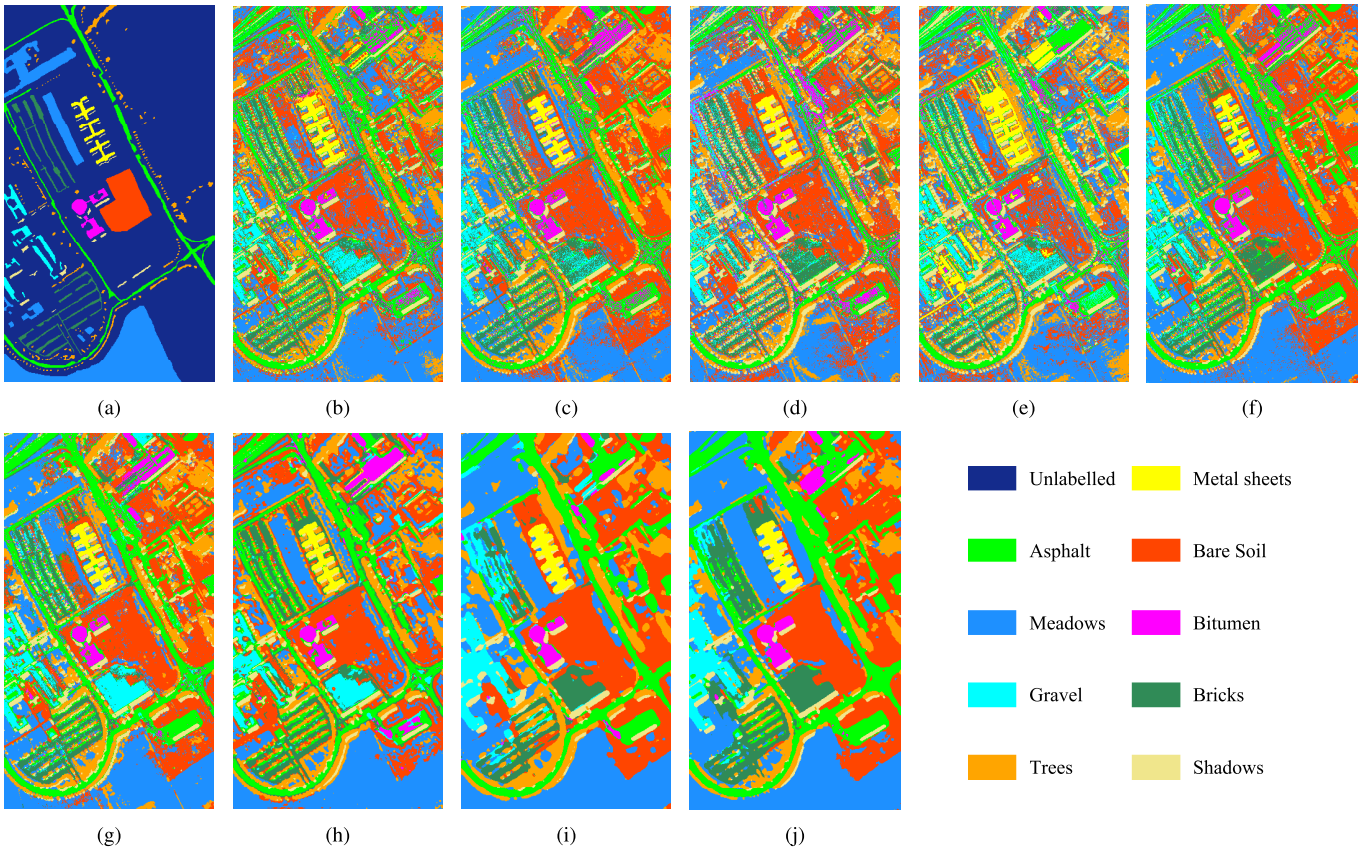


Fig. 5. Classification maps for the PaviaU image. (a) Ground truth. (b) LCNN (OA = 80.85). (c) DCNN (OA = 82.56). (d) DGAN (OA = 82.48). (e) CRNN (OA = 73.93). (f) MCNN (OA = 89.36). (g) PPCNN (OA = 90.36). (h) NLGCN (OA = 94.73). (i) ANNA (OA = 95.39). (j) MCNNS (OA =97.75).

The parameters of the reference methods are set to the default values indicated in their original works. We also compare the proposed method to its reduced version: MCNNS without involving the spatial postprocessing (MCNN), to verify the effectiveness of its spectral feature learning capability.

Tables II and III report the class-specific accuracy, AA, OA, and κ of the tested methods on the PaviaU and Salinas

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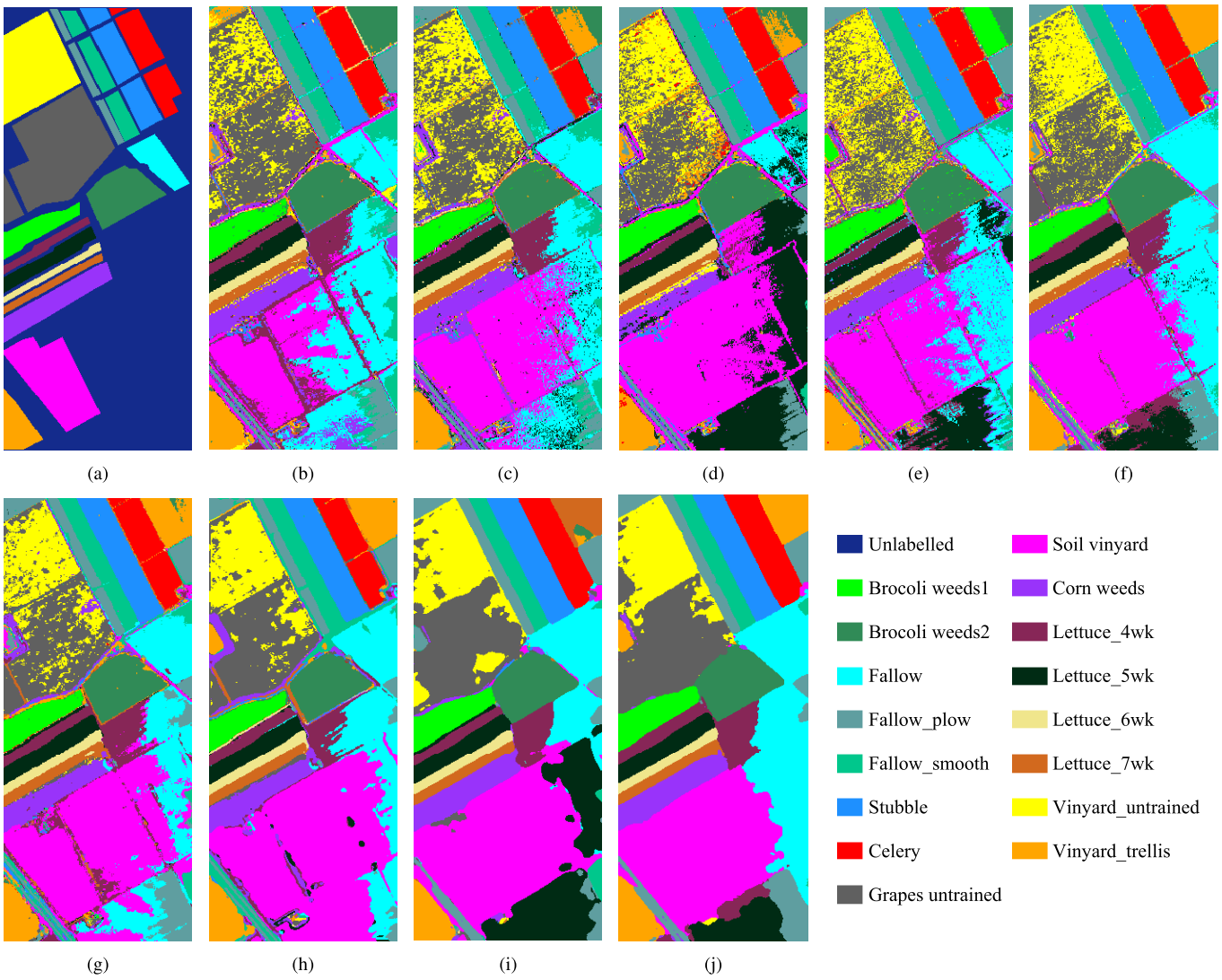


Fig. 6. Classification maps for the Salinas image. (a) Ground truth. (b) LCNN (OA = 86.54). (c) DCNN (OA = 86.93). (d) DGAN (OA = 84.97). (e) CRNN (OA = 84.81). (f) MCNN (OA = 91.23). (g) PPCNN (OA = 91.22). (h) NLGCN (OA = 93.59). (i) ANNA (OA = 95.17). (j) MCNNS (OA = 96.25).

datasets. As can be observed, the proposed MCNNS consistently yields the best AA, OA, and κ with a favorable improvement over the reference methods for the two datasets. For example, on PaviaU (Table II), the improvement in OA compared to LCNN, DCNN, DGAN, CRNN, PPCNN, NLGCN, and ANNA methods is about 16.7%, 15.2%, 15.3%, 24.8%, 7.0%, 3.4%, and 2.2%, respectively. For the comparison of spectral classifiers, the proposed MCNN improves significantly the classification performance compared to CNN-based spectral classifiers (LCNN and DCNN), GAN-based spectral classifier (DGAN), and RNN-based spectral classifier (CRNN) on the two images. The gains in OA compared to the best spectral classifiers are about 7.3% and 3.6% for the PaviaU and Salinas images, respectively. This indicates that the proposed MCNN has powerful spectral feature learning capability.

We further test the proposed method on two challenging HSIs: Houston2013 and Houston2018. We did not report the results of DGAN, NLGCN, and ANNA on the two images due to their high memory requirements. Instead, we add the results of two related methods: GRNN [35] and CAE [58]. Tables IV and V report the class-specific accuracy, AA, OA, and κ of the tested methods on the two datasets. The proposed MCNNS consistently yields the best OA and κ with a significant improvement over the reference methods for the two datasets. For instance, in Table V, the proposed method achieves OA 70.25%, with the gains of 17.5%, 17.3%, 17.1%, 21.5%, 13.8%, and 4.6% over LCNN, DCNN, GRNN, CRNN, PPCNN, and CAE methods, respectively. It is also evident that the proposed MCNN improves significantly the classification performance compared to both CNN-based spectral classifiers (LCNN and DCNN) and RNN-based spectral classifiers (GRNN and CRNN) on the two images. The improvements in OA compared to the best spectral classifiers are approximately 8.5% and 12.2% for the Houston2013 and Houston2018 images, respectively. It is worth highlighting that our MCNN (spectral classification alone) performs comparable in Houston2018 and even performs better in Houston2013 than the PPCNN and CAE methods (that use spatial information) in

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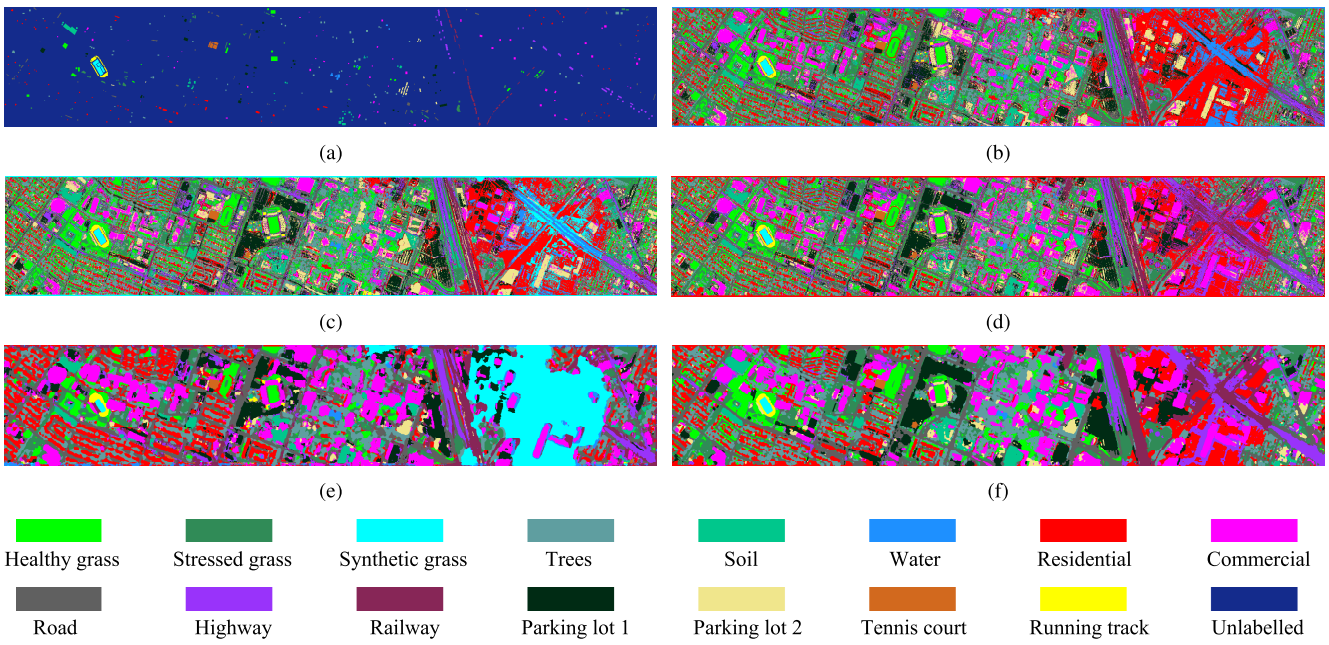


Fig. 7. Classification maps for the Houston2013 image. (a) Ground truth. (b) DCNN (OA = 77.76). (c) GRNN (OA = 79.06). (d) MCNN (OA = 87.72). (e) CAE (OA = 79.42). (f) MCNNS (OA = 89.31).

terms of classification performance. It can be concluded that the proposed method without overlapping problems yields the best classification accuracy compared to the reference methods for all four HSIs. In terms of the class-specific accuracy, our method yields worse results compared to the best ones in most of the classes on Houston2018 (leading to a smaller AA compared to CAE), but our method yield better OA and κ in this case as well. The main reason is that many classes (e.g., "Residential," "Commercial," "Road," and "Sidewalk") in Houston2018 have huge spectral similarity together with less spectral bands, leading our method to a relatively weaker spectral discriminative capability.

Apart from quantitative analysis, Figs. 5–8 show the full classification maps obtained by different methods on the four images. Visually, they are consistent with the results reported in Tables II–V. Obviously, the spectral classifiers (LCNN, DCNN, DGAN, GRNN, CRNN, and MCNN) often exhibit noisier estimations than spectral–spatial classifiers (NLGCN and CAE) and spatial postprocessing methods (PPCNN, ANNA, and MCNNS), mainly because of spectral variability and mixture pixels. Among them, the proposed MCNNS presents more similar results to the reference maps than all the reference methods. It is worth highlighting that the proposed MCNN shows appealing estimations on two challenging images (Houston2013 and Houston2018) compared to all the reference methods, benefiting from its discriminative spectral feature representation capability.

We also compare the proposed MCNN with three deep spectral classifiers: miniGCN [37], SpeFormer [38], and TCNN [40]. We further compare the proposed MCNNS with two spectral classifiers with spatial postprocessing: CMRF [56] and TCNNS [40], as well as three spectral–spatial classifiers: SpaFormer [38], FCGCN [37], and MCNCP [60]. The comparison results are summarized in Table VI. We did not report the results of CMRF on Houston2013 and Houston2018 as well as the results of miniGCN and FGCNN on Houston2018 due to their high memory requirements. The results in Table VI show that our MCNN consistently yields the best OA over the reference spectral classifiers (i.e., miniGCN [37], SpeFormer [38], and TCNN [40]) for all four datasets. Also, our MCNNS yields better (except TCNNS on Salinas) OA than the reference methods for all four datasets.

C. Hyperparameter Analysis

1) Analysis of Different Volume of Training Data: We analyze the influence of the volume of training data on the classification accuracy of the proposed method. For the Houston2013 and Houston2018 images, we randomly select different percentages of the training set for training. The results in Fig. 9 show that the OA generally improves when the volume of training data increases for all four datasets. In particular, on PaviaU and Salinas, the proposed method achieves OA of over 82% with only ten training samples per class, which demonstrates the advantage of the proposed method under limited training data.

2) Analysis of the Multiview Spectral Feature Learning: To validate the proposed multiview spectral feature learning architecture, we compare it with the architectures that contain one of the views ($P = 1, 3,$ and 5) and two of the three views ($P = 1\&3, 1\&5,$ and $3\&5$). The results in Table VII show that the proposed multiview ($P = 1\&3\&5$) consistently yields better accuracy than any of its single- and two-view architectures. It is also evident that each single view of our architectures performs better than the reference spectral classifiers (LCNN, DCNN, DGAN/GRNN, and CRNN) on four datasets (except $P = 3$ on PaviaU, see Tables II–V), and each two-view of our architectures performs better than single-view architecture

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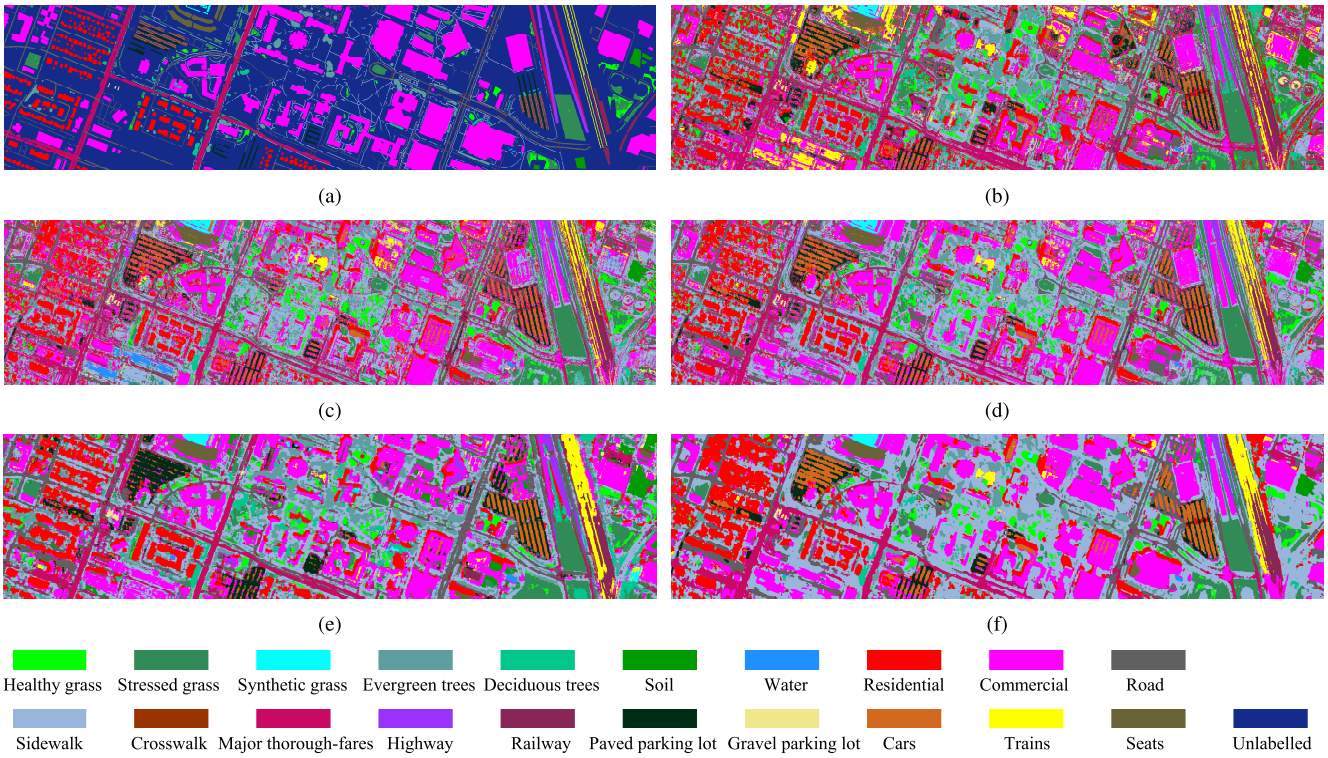


Fig. 8. Classification maps for the Houston2018 image. (a) Ground truth. (b) DCNN (OA = 53.37). (c) GRNN (OA = 53.67). (d) MCNN (OA = 65.84). (e) CAE (OA = 64.43). (f) MCNNS (OA = 70.69).

TABLE VI

OA OBTAINED BY SEVERAL SPECTRAL CLASSIFIERS (MINIGCN, SPEFORMER, TCNN, AND OUR MCNN), SPECTRAL CLASSIFIERS WITH SPATIAL POSTPROCESSING (CMRF, TCNNS, AND OUR MCNNS), AND SPECTRAL–SPATIAL CLASSIFIERS (SPAFORMER, FGCNN, AND MCNCP)

Image	miniGCN	SpeFormer	TCNN	**MCNN**	CMRF	TCNNS	SpaFormer	FGCNN	MCNCP	**MCNNS**
PaviaU	84.24	84.00	88.80	89.20	88.99	97.04	89.83	91.87	93.11	97.09
Salinas	88.32	87.16	90.46	90.50	91.66	95.52	91.76	91.86	95.04	95.28
Houston2013	78.91	79.36	81.15	87.44	-	82.74	82.20	80.40	83.04	89.07
Houston2018	-	57.99	58.84	65.38	-	64.91	62.02	-	65.97	70.25

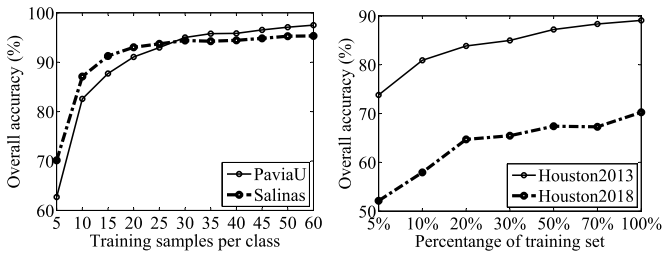


Fig. 9. OA of the proposed TCNNS with different volumes of training data in four datasets. (a) PaviaU and Salinas. (b) Houston2013 and Houston2018.

TABLE VII

EFFECT OF THE NUMBER OF SPECTRUM INPUTS ON OA FOR FOUR HSIS

Input	PaviaU	Salinas	Houston2013	Houston2018
$P = 1$	87.17±1.42	90.22±0.57	86.83±0.92	61.81±1.31
$P = 3$	80.35±2.57	89.65±1.08	81.88±0.30	59.07±1.68
$P = 5$	87.16±1.44	90.07±1.15	82.83±0.38	60.08±1.77
$P = 1\&3$	88.31±1.50	90.17±0.98	85.75±0.88	64.59±0.88
$P = 1\&5$	88.75±1.35	90.25±0.91	86.36±0.43	64.77±0.60
$P = 3\&5$	89.06±1.64	90.20±1.04	85.22±0.75	65.19±1.09
$P = 1\&3\&5$	89.20±1.50	90.50±0.74	87.44±0.76	65.38±1.56

(except $P = 1$ on Salinas and Houston2013) and the reference spectral classifiers. This demonstrates the effectiveness of the proposed multiview spectral feature learning framework.

We further analyze the effect of the proposed two local streams (i.e., $P = 3\&5$) with different index orderings: RCD (i.e., the row axis first and then the column axis to the depth axis), CRD, DRC, and DCR. We did not include $P = 1$ because it is the same for different index orderings. The results in Fig. 10 show that RCD and CRD perform better than DRC and DCR on OA for all the datasets. The reason is that RCD or CRD provides the spectral context in the first two dimensions, which allows the proposed network to extract more discriminative local spectral contextual features.

3) Analysis of the Interactive Attention: We compare the proposed interactive attention with the versions: without any attention (None), attention without using interaction (Single), our attention embedding in all the triple streams (All-stream),

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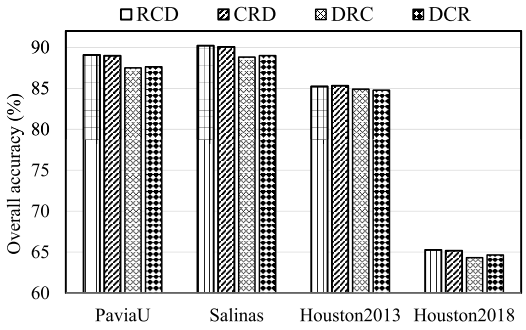


Fig. 10. Effect of different index orderings for 3-D spectrum construction on OA in four datasets.

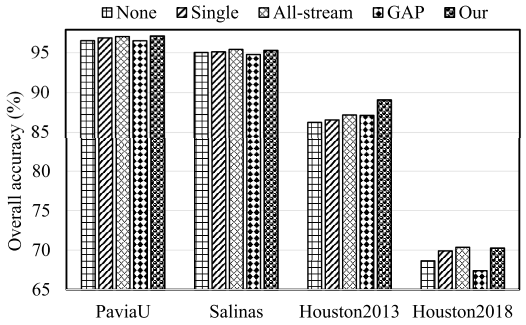


Fig. 11. Effect of different attention schemes on the OA in four datasets.

TABLE VIII

COMPARISON OF THE PROCESSING TIME AND THE NUMBER OF PARAMETERS ON DIFFERENT METHODS FOR THE PAVIAU IMAGE

Method	DCNN	CRNN	PPCNN	CAE	MCNNS
Training (s)	40.1	46.08	786.5	143.7	163.2
Testing (s)	4.2	8.5	26.9	10.1	10.8
#Params ($\times$M)	0.05	0.35	0.10	0.37	8.58

and our attention using GAP. The results in Fig. 11 show that our interactive attention performs more stable and better or comparable OA compared to different versions of attention for all the test datasets. This is because our attention preserves more detailed contextual features and learns their interactive correlations.

4) Analysis of the Computational Efficiency: The results in Table VIII provide comparative analysis with respect to the training and testing time and the number of parameters for the proposed MCNNS and four representative reference methods: DCNN [33], CRNN [36], PPCNN [34], and CAE [58]. The reported values correspond to the PaviaU image and are similar to the other images. In the training and testing procedures, our MCNNS is more time-consuming compared to the reference methods (except PPCNN). The proposed method involves much more parameters than all the reference methods due to a wider triple-stream network with interactive attention, while we obtain better results in terms of classification accuracy.

V. CONCLUSION

In this article, we proposed a unified multiview spectral feature learning framework based on 2-D CNN for HSI classification. The proposed framework simultaneously captures the discriminative global and local spectral contextual features via a multiview spectrum construction method with a triple-stream feature learning network. Moreover, we devised an interactive attention method embedding into the framework to further enhance the spectral feature learning capability based on global correlations. This framework improves significantly the spectral classification performance, especially for challenging HSIs. In addition, we developed a decision fusion method to effectively integrate the global and local spectral features of the triple streams. Experimental results on four hyperspectral datasets demonstrated the state-of-the-art classification performance.

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